

Faithful by Construction: Claim-Anchored Attribution for Multi-Document Summarization

Shuo Guan

UBS AG

NY, 10010

shuo.guan@ubs.com

Abstract

End-to-end large language models (LLMs) produce fluent multi-document summaries but remain prone to hallucination, and the attributions they offer are typically coarse (whole documents or passages) and generated post hoc, leaving each summary statement hard to verify. We revisit the modular Extract–Select–Rewrite paradigm and recast its intermediate representation as the unit of attribution. We present CAMS, a Claim-Anchored Multi-document Summarization framework that (i) extracts atomic claims with token-level provenance from every source document, (ii) clusters equivalent claims across documents while flagging inter-source conflicts, (iii) selects a support-aware and salient subset, and (iv) rewrites the selection into a summary in which every sentence is anchored to a support-checked claim that links back to one or more source spans. Because content is localized before it is realized, the pipeline is attribution-oriented by construction and faithfulness-oriented by construction: it structurally preserves fine-grained, multi-source traceability while using support-aware selection, constrained rewriting, and verification to encourage, rather than guarantee, factual faithfulness. We evaluate quality, faithfulness, and localization on MultiNews, analyze conflict handling on DiverseSumm, and test zero-shot transfer on WCEP, using a two-regime protocol that separates reference-free citation quality from gold-aligned localization accuracy, and we add an evaluator-decoupled audit that tests citation precision with a support model never used for selection or verification. CAMS matches strong end-to-end and span-attribution baselines on summary quality while substantially improving faithfulness and citation precision, lifting multi-source attribution accuracy by roughly two-thirds, and exposing a controllable faithfulness–coverage trade-off that end-to-end models leave implicit.

1 Introduction

Multi-document summarization (MDS) condenses a cluster of topically related documents into a short, coherent summary (Fabbri et al., 2019). Modern LLMs are strong abstractive summarizers, yet two problems persist that are especially acute in the multi-source setting. First, *faithfulness*: abstractive systems can hallucinate content unsupported by—or contradicting—their inputs (Maynez et al., 2020; Goyal and Durrett, 2022). Second, *verifiability*: even when systems cite their sources, the citations are usually coarse—pointing to entire documents or paragraphs—and are produced after generation, so a reader must still sift through irrelevant text to confirm any given statement (Gao et al., 2023; Slobodkin et al., 2024).

MDS compounds both problems. The same fact recurs across sources (redundancy), different sources disagree (conflict), and a single summary statement is often supported by evidence drawn from several documents at once (multi-source support). End-to-end models tend to flatten these phenomena: they silently deduplicate, paper over disagreement, and emit attributions that point to a single best document even when support is distributed (Huang et al., 2024).

We argue that a modular pipeline with a structured intermediate representation is uniquely suited to these challenges. Prior work shows that augmenting abstractive summarization with explicit structure—linearized knowledge structures and salient keyword topics—improves content quality (Guan, 2021); we push this further by making the structured unit itself the anchor of attribution and verification. Building on the Extract–Select–Rewrite paradigm (Guan and Padmakumar, 2023), we recast its triple-like intermediate as the unit of both verification and attribution. Our framework, CAMS, decomposes summarization into (1) atomic-claim extraction with token-level prove-

nance, (2) cross-document claim clustering with conflict detection, (3) faithfulness- and salience-aware content selection, and (4) attributable rewriting in which each generated sentence carries pointers back to the claim—and hence the source spans—it realizes. A lightweight verification step rejects rewrites that drift from their selected claims.

The pipeline is thus *attribution-oriented by construction*: every generated sentence must resolve to selected claim identifiers, and those identifiers map deterministically back to source spans. Faithfulness, by contrast, is treated as an *objective* rather than a guaranteed invariant—encouraged by self-support scoring, bidirectional entailment checks during clustering, and post-rewrite verification—because each learned step can still fail: an extractor may misread a passage, decontextualization may restore the wrong entity, quote matching shows only that a quote occurs in the source, clustering may merge non-equivalent claims, and NLI verifiers can misclassify support. We therefore treat provenance as a structural invariant but faithfulness as a testable objective, audited with support models separated from selection and verification. Unlike span-first attribution (Slobodkin et al., 2024), which anchors generation to raw spans, our claims are normalized, recombinable units that merge across documents and are checked individually, bridging citation-style attribution (Gao et al., 2023; Rashkin et al., 2023) with atomic-fact factuality evaluation (Min et al., 2023).

Our contributions are: (i) a claim-anchored modular MDS framework whose intermediate representation serves simultaneously as attribution anchor and verification unit; (ii) explicit cross-document claim clustering and conflict handling that enable multi-source, span-level citations; (iii) a content-selection interface that turns the faithfulness–coverage trade-off into a controllable knob; and (iv) a two-regime evaluation protocol that cleanly separates citation quality from localization accuracy, including model-decoupled citation checks and a multi-source attribution metric designed to test whether structured claims improve over span-first baselines.

2 Related Work

Faithful summarization and its evaluation. Faithfulness in abstractive summarization has been studied extensively (Maynez et al., 2020; Kryscinski et al., 2020). Evaluation has moved from *n*-

gram overlap (Lin, 2004) toward entailment- and QA-based consistency metrics (Laban et al., 2022; Fabbri et al., 2022; Zha et al., 2023; Honovich et al., 2022) and, most recently, decomposition into atomic facts (Min et al., 2023). Our intermediate representation aligns naturally with atomic-fact evaluation: the unit we generate is the unit these metrics score.

Attributed text generation. A growing line of work asks models to emit text together with supporting evidence (Rashkin et al., 2023; Bohnet et al., 2022). ALCE (Gao et al., 2023) introduced a reproducible benchmark with NLI-based citation precision and recall, but most systems cite at passage granularity after generating. Closest to us, “Attribute First, then Generate” (Slobodkin et al., 2024) selects source spans before generating and reuses them as attributions. We differ in the type of intermediate: normalized atomic claims rather than raw spans, which lets us merge equivalent evidence across documents and produce multi-source citations—central to MDS—and to verify each claim independently.

Modular summarization. Pipeline approaches trade some end-to-end performance for modularity, interpretability, and controllability. Related work injects structured signals—linearized knowledge structures and keyword topics—into abstractive sentence summarization to improve salience and quality (Guan, 2021). The Extract–Select–Rewrite method (Guan and Padmakumar, 2023) extracts relation triples, selects a subset, and realizes them with a fine-tuned rewriter. We extend this paradigm to the multi-document setting, replace brittle off-the-shelf extraction with LLM-based atomic-claim decomposition carrying provenance, and add cross-document clustering, conflict detection, and a verification loop.

Multi-document summarization. MDS requires modeling cross-document relations, redundancy, and coverage (Fabbri et al., 2019; Xiao et al., 2022). Classical systems balance relevance and redundancy with criteria such as MMR (Carbonell and Goldstein, 1998); recent benchmarks stress diverse and conflicting information across sources (Huang et al., 2024). Our clustering and conflict-detection modules address these directly and expose them in the attribution layer.

3 Method

3.1 Overview and Notation

Given a cluster of source documents $D = \{d_1, \dots, d_K\}$, CAMS produces a summary Y together with, for each summary sentence, a set of supporting source spans. We write a token span as $c = (\text{doc}, \text{start}, \text{end})$. The pipeline has five stages (Figure 1): claim extraction, cross-document clustering and conflict detection, content selection, attributable rewriting, and verification.

3.2 Claim Extraction with Provenance

For each document d_k we prompt an LLM to decompose its content into a set of atomic claims—minimal, self-contained propositions (Min et al., 2023). A direct demand for character or token offsets is unreliable: instruction-tuned models routinely miscount positions and hallucinate spans. We therefore decouple *what* is claimed from *where* it is stated. The extractor returns, for each claim, a structured record $\langle t_i, q_i, k \rangle$ comprising the (decontextualized) claim text t_i , a *verbatim* quotation q_i copied from d_k that licenses the claim, and the document id k ; it never reports offsets. Provenance is then *resolved* deterministically by locating q_i in d_k , yielding the token span $c_i = (k, \text{start}, \text{end})$ and the candidate set $C_k = \{(t_i, c_i)\}$. This resolves the quote’s location but does not by itself guarantee that t_i is entailed by q_i ; that support is scored during selection and rechecked after rewriting. Unlike off-the-shelf OpenIE (Angeli et al., 2015) used in prior modular work (Guan and Padmakumar, 2023), LLM decomposition improves recall and yields fluent, normalized claims while still anchoring each to its source at the first stage rather than recovering it later.

Decontextualization. Atomic claims must stand alone outside their paragraph, so the extraction prompt requires *decontextualized* propositions (Choi et al., 2021): pronouns and bridging references are resolved, and elided subjects, times, and locations are restored (e.g., “he said troops would be sent” becomes “Defense Minister X said on Tuesday that troops would be sent”). The instruction is issued inline during extraction; crucially, the verbatim quote q_i is stored separately so that rewriting the claim for self-containedness never corrupts the string used for localization. Because decontextualization can introduce wrong fillers, claim and quote are kept as separate, separately support-

checked fields rather than treated as equivalent.

Long documents. When d_k exceeds the model context window we split it into overlapping chunks at paragraph boundaries with a sliding window, extract per chunk, and add each chunk’s global character offset back to the resolved spans before merging, so provenance is expressed in the coordinate system of the full document. Claims duplicated across the overlap region are removed at the clustering stage (§3.3).

Quote-to-span resolution. Given q_i we locate it in d_k with a three-tier matcher (Algorithm 1, Appendix C). We first attempt an exact substring match. On failure—typically minor LLM normalization such as smart quotes, collapsed whitespace, or a dropped article—we fall back to approximate matching over a normalized form (case-, whitespace-, and punctuation-normalized) using a windowed similarity search (rapidfuzz/difflib), accepting the best window whose normalized indel similarity exceeds ρ . The matched character interval is finally converted to a token span through the tokenizer’s `offset_mapping`, giving $(k, \text{start}, \text{end})$ on the same token grid used by the localization metrics (§4.3). Claims whose quote cannot be matched above ρ are dropped, keeping provenance precise at a small cost to recall.

3.3 Cross-Document Clustering and Conflict Detection

The union $C = \bigcup_k C_k$ contains many near-duplicate claims. We embed claim texts with a sentence bi-encoder (Li et al., 2023) and group them by agglomerative clustering under a cosine-distance threshold, so that semantically equivalent claims merge into a cluster g whose provenance is the union of its members’ spans, $\text{SPANS}(g) = \bigcup_{i \in g} \{c_i\}$, possibly spanning multiple documents. We use *average* linkage (complete behaves near-identically; single linkage is avoided for its chaining). Because embedding proximity conflates equivalence with relatedness—and even contradiction, since “30 killed” and “47 killed” sit close—we confirm each merge with *bidirectional* entailment from a three-class NLI model, refusing non-mutually-entailing merges. This guards against mis-merges rather than proving equivalence, so clustering-sensitive multi-source attribution is evaluated separately (§4.3, §5.4).

Conflict detection. Running NLI over all $\binom{|C|}{2}$ pairs is quadratic and wasteful. We instead build a candidate pool: cross-cluster pairs that are embedding-similar yet were *not* merged, plus pairs whose claims mention the same named entity or numeric quantity (cheap NER and number matching). Only candidate pairs are scored by the three-class NLI model; a pair labeled CONTRADICTION in either direction above a confidence threshold induces a conflict link between its clusters (e.g., differing casualty counts). Clustering yields the deduplicated, multi-source evidence units that make distributed attribution possible; conflict links let later stages surface disagreement rather than silently resolve it.

Support-model separation. Detecting disagreement requires separating CONTRADICTION from mere non-entailment, so conflict detection uses a genuine *three-class* NLI model (entailment/neutral/contradiction). Support decisions are deliberately split across three frozen models to avoid circular evaluation. The selector uses a lightweight TRUE-style model M_{sel} only to score whether an extracted claim is backed by its own spans. The verifier uses a separately calibrated TRUE-style model M_{ver} for generation-time repair and rejection (§3.6). Reported citation precision, however, is not computed with either of these models: Regime 1 uses an independent support evaluator M_{eval} (a SummaC-style NLI evaluator, Laban et al., 2022) that is never used in extraction, selection, rewriting, verification, threshold tuning, or repair (§4.3). We still report the original TRUE-style score as a compatibility diagnostic, but the main precision claim is based on M_{eval} .

3.4 Content Selection

The selector chooses which clusters enter the summary by trading off *salience*, *self-support*, and *coverage*. The first two are *pointwise* and are produced by a gradient-boosted decision tree (GBDT) (Ke et al., 2017; Friedman, 2001) that scores each cluster g from cheap, interpretable features: salience features include the cluster size and the number of distinct source documents it spans, encoding the multi-document intuition that cross-source redundancy signals importance; the self-support feature is M_{sel} ’s entailment score of the claim text against its own spans $\text{SPANS}(g)$, penalizing clusters whose own evidence does not back them. This feature is used only before generation and is never used to

compute reported citation precision; Appendix D further ablates it to check that results are not an artifact of optimizing a single evaluator. Choosing a GBDT over a neural scorer is deliberate: although neural classifiers such as CNNs are effective given abundant labeled data (Wei et al., 2018), a GBDT is far more sample-efficient on a few thousand examples, which underlies the graceful low-resource degradation in §5.9.

Coverage is sequential. Coverage—marginal new information in the MMR sense (Carbonell and Goldstein, 1998)—depends on what has already been selected and so cannot be a pointwise feature. Selection is therefore a greedy loop (Algorithm 2, Appendix C): with S the set chosen so far, each remaining cluster is scored

$$\text{score}(g \mid S) = \underbrace{s_{\text{GBDT}}(g)}_{\text{salience+self-support}} - \lambda \max_{g' \in S} \text{sim}(g, g'),$$

the highest-scoring cluster above admission threshold θ is added to S , and the loop repeats until no cluster qualifies. The GBDT supplies only the pointwise term s_{GBDT} ; the redundancy penalty is recomputed against S at every iteration. The threshold θ exposes an explicit *faithfulness–coverage trade-off*: lowering it admits more clusters at some risk to precision, raising it yields conservative, highly supported summaries; we sweep $\theta \in [0.3, 0.8]$ (§5.7). For conflicting clusters, a policy decides whether to surface both sides (attributing each to its sources) or to defer; we treat this as an ablatable design choice (§4).

Training labels by distant supervision. We obtain silver labels without per-cluster annotation (Mintz et al., 2009): each gold reference sentence is aligned to clusters by M_{sel} entailment (backing off to embedding/ROUGE similarity when borderline), and clusters covered by some reference sentence are positive, the rest negative. Crucially, neither M_{ver} nor M_{eval} is used to create these labels, so the selector is trainable independently of the other modules, preserving the low-resource and style-control benefits of modular design (Guan and Padmakumar, 2023).

3.5 Attributable Rewriting

The rewriter realizes the selected clusters into fluent summary sentences while preserving their provenance. We give each selected cluster a short identifier g_j and present the rewriter with the labeled

claim list (“[g3]... [g7]...”), instructing it to terminate every sentence with an inline citation marker, e.g. [g3, g7]. We parse these markers and resolve each g_j through the stored map $g_j \mapsto \text{SPANS}(g_j)$, so each sentence inherits span-level—and, when it cites clusters from different documents, multi-source—citations with no separate attribution step. To aid coherence we pre-order the selected clusters (chronologically for event-centric inputs, by salience otherwise) and let the rewriter reorder locally; conditioning only on selected claim texts limits reintroduction of unselected content, and residual drift is caught by verification.

3.6 Verification

A verification module checks each generated sentence in two directions against the spans it cites. The *support* (recall) direction tests whether the cited spans entail the sentence, requiring M_{ver} support above τ_v . The *no-overflow* (precision) direction decomposes the sentence into atomic facts and checks that *each* is entailed by the cited spans, in the spirit of FActScore (Min et al., 2023) and again using M_{ver} rather than the final evaluator; this catches the extra unsupported clause a sentence-level check can miss. A sentence passes only if both succeed. On failure we apply at most R repair retries: if the sentence is well-formed but cites the wrong cluster, we re-retrieve the best supporting cluster by NLI and *repair the citation*; otherwise we re-decode from the same claim set. Sentences still failing are dropped. A verifier pass is an automatic support judgment, not a proof of truth, which is why the final audit uses a model never invoked by the selector or verifier.

4 Experimental Setup

4.1 Datasets

We evaluate on three news-domain MDS datasets. MultiNews (Fabbri et al., 2019) is the standard large-scale benchmark; we additionally use its human sub-sentential source–summary span alignments (Ernst et al., 2021) as gold for fine-grained localization (§4.3). DiverseSumm (Huang et al., 2024) targets diverse and conflicting information across sources and supplies human-validated references, supporting our conflict-handling study. WCEP (Gholipour Ghalandari et al., 2020) provides short, event-centric summaries; we use it as a *zero-shot transfer* setting, applying the

MultiNews-trained selector without any WCEP-specific training (§5.6). In short, MultiNews is our primary benchmark; DiverseSumm is used for conflict/diversity analysis; WCEP is used only as an event-centric zero-shot transfer setting. Dataset statistics appear in Appendix A.

4.2 Baselines

We compare against: (1) a strong end-to-end LLM prompted zero/few-shot to summarize and to emit passage-level citations, in the ALCE style (Gao et al., 2023); (2) Attribute First, then Generate (Slobodkin et al., 2024), the closest span-attribution method, in the same MDS setting; (3) a fine-tuned MDS model (PRIMERA (Xiao et al., 2022) / LED (Beltagy et al., 2020)) paired with a post-hoc fact-checker; and (4) the original ESR (Guan and Padmakumar, 2023) naively applied to concatenated documents, as an ablation isolating the contribution of cross-document clustering.

4.3 Evaluation Protocol

We report summary quality with ROUGE (Lin, 2004) and BERTScore (Zhang et al., 2020), plus an LLM-as-judge assessment of coherence and coverage (Liu et al., 2023; Zheng et al., 2023); faithfulness with AlignScore (Zha et al., 2023), SummaC (Laban et al., 2022), and atomic-level FActScore (Min et al., 2023). Unless stated, MultiNews uses the full 5,622-example test split (diagnostic runs use marked 500/1,000-example subsets); DiverseSumm uses all 245 examples and WCEP its $\sim 1,000$ -example public test split. For attribution we use a two-regime protocol.

Regime 1 (citation quality, reference-free). Following ALCE (Gao et al., 2023), we compute NLI-based citation recall and precision over the actual generated summaries, adapted to multi-source spans, together with citation conciseness (mean cited tokens and spans per claim). To address evaluator circularity, the primary Regime 1 numbers are produced by M_{eval} , an independent support evaluator that is never used by CAMS for selection, verification, repair, or threshold tuning. We additionally report the original TRUE-style precision as a diagnostic for comparability with prior work; conclusions must hold under both evaluators. All systems’ citations are first projected to a common token-span representation so comparisons are like-for-like.

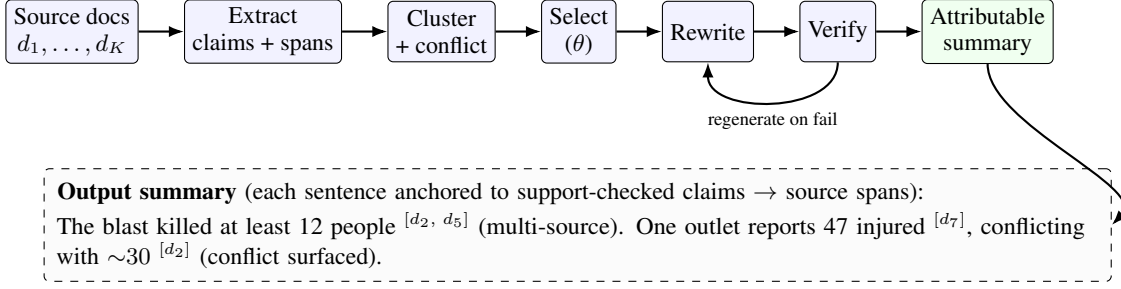


Figure 1: Overview of CAMS. Documents are decomposed into atomic claims with token-level provenance; equivalent claims are clustered across documents (with conflicts detected), a support-aware and salient subset is selected, and the selection is rewritten so that each summary sentence carries citations to one or more source spans. The output card illustrates the three MDS-specific properties: multi-source attribution, span-level citations, and surfaced conflict.

Regime 2 (localization accuracy, gold-aligned).

To isolate localization from generation noise, we feed gold reference propositions as oracle content and measure whether each system points to the correct source spans, scored against the human alignments via soft (token-IoU $\geq \tau$) span precision/recall/F1 and a multi-source recall metric restricted to propositions whose gold support spans ≥ 2 documents. The main localization threshold is $\tau=0.5$, with sensitivity at $\tau \in \{0.3, 0.5, 0.7\}$. We additionally report a human study on a 100–300-claim sample per system, using three annotators to label per-claim support and measuring verification time; agreement is summarized with Cohen’s or Fleiss’ κ , depending on whether a pairwise or multi-rater analysis is reported.

4.4 Implementation Details

Claim extraction (with inline decontextualization), atomic-fact decomposition, and rewriting all use the same frozen claude-opus-4-8 model with three few-shot exemplars; the LLM baselines share it for a controlled comparison. Claim texts are embedded with gte-large (Li et al., 2023) and clustered by agglomerative average-linkage at cosine similarity 0.82, with merges confirmed by bidirectional entailment. Conflict detection uses a three-class DeBERTa-v3 (He et al., 2023) model fine-tuned on MNLI (Williams et al., 2018) (contradiction threshold 0.7); selection, verification, and citation evaluation use *distinct* TRUE-/SummaC-style models M_{sel} , M_{ver} , and M_{eval} (Honovich et al., 2022; Laban et al., 2022), with M_{eval} never queried during generation, filtering, threshold tuning, or repair. Documents are chunked into 512-token windows (64-token overlap); quotes are resolved by exact then rapidfuzz indel matching

($\rho=0.85$) before conversion to token spans. The pointwise selector is a LightGBM GBDT (Ke et al., 2017) trained on distant-supervision labels (§3.4); the MMR weight is $\lambda=0.5$ and the main operating point $\theta^*=0.6$ (swept over $\{0.3 \dots 0.8\}$). Verification allows two repair retries (citation repair, then regeneration, then drop). Experiments run on $4 \times$ A100 GPUs; full hyperparameters, prompts, schemas, and per-module algorithms are in Appendices A, B, and C (Table 7).

5 Results and Analysis

Unless noted, faithfulness and attribution gains over every baseline are statistically significant (paired bootstrap, $p < 0.05$); ROUGE is reported for reference only, since it correlates weakly with faithfulness (Maynez et al., 2020; Goyal and Durrett, 2022). Per-dataset statistics, full ROUGE, τ -sensitivity, confidence intervals, and per-module diagnostics are reported in the Appendix.

5.1 Main Results: Quality and Faithfulness

Table 1 reports quality and faithfulness on Multi-News. CAMS is competitive with the strongest fine-tuned summarizer on surface quality—within 0.4 ROUGE-L of PRIMERA and best on BERTScore—while improving faithfulness across AlignScore, SummaC, and FActScore over every baseline. Removing the verification loop (– verification) lowers faithfulness without much affecting quality, indicating that part of the gain comes from verification rather than the rewriter alone.

5.2 Citation Quality (Regime 1)

Table 2 reports citation quality on the generated summaries under the independent evaluator M_{eval} , with the original TRUE-style precision shown only

System	R-L	BERT Sc.	Align Sc.	Sum maC	FAct Sc.
E2E LLM (ALCE)	22.8	86.5	.79	.71	.75
Attribute First	23.9	87.0	.85	.79	.82
PRIMERA + chk.	25.4	87.4	.81	.74	.78
ESR (concat)	22.0	85.8	.83	.78	.80
CAMS (ours)	25.0	87.6	.91	.87	.90
– clustering	24.1	87.1	.89	.85	.88
– verification	24.8	87.4	.86	.82	.85

Table 1: Quality and faithfulness on MultiNews. ROUGE-1/2 and additional implementation details are reported in Appendix A.

System	Cite Rec (M_{eval})	Cite Prec (M_{eval})	TRUE Prec diag.	Tok/claim	Span/claim
E2E LLM (ALCE)	70	59	61	95	1.0
Attribute First	78	76	78	34	1.6
CAMS (ours)	81	84	88	22	1.8

Table 2: Regime 1 citation quality, reference-free. Recall and precision are scored by the independent evaluator M_{eval} ; TRUE-style precision is reported only as a compatibility diagnostic. Tok/Span per claim measure conciseness (lower is better).

as a diagnostic. CAMS still attains the highest citation precision under the independent evaluator (84), but the margin is smaller than under the internal TRUE-style diagnostic (88). This is the desired sanity check: the gain is not reported solely on the model family used to filter CAMS outputs. CAMS also remains the most concise system (22 tokens per claim) at recall comparable to or above the baselines. It cites slightly more spans per claim than Attribute First (1.8 vs 1.6)—a direct consequence of aggregating evidence across documents—yet uses far fewer total tokens, because each cited span is sub-sentential rather than a whole passage.

Circularity audit. Because CAMS explicitly filters and repairs generated sentences with M_{ver} , a high TRUE-style precision score alone would not be sufficient evidence: it could reflect optimization toward the verifier rather than genuinely better citations. We therefore treat the independent M_{eval} score and human precision correlation as the primary validity checks. The ranking is stable under M_{eval} , but the CAMS precision advantage shrinks from 10 points under the TRUE-style diagnostic (88 vs. 78) to 8 points under the independent evaluator (84 vs. 76), which is consistent with a real but more conservative gain. Appendix D further reports that removing the self-support feature or

the verification loop reduces precision under both evaluators, suggesting that the modules improve support rather than merely exploiting one scoring function.

5.3 Fair Post-Processing Control

To check that CAMS’s precision advantage is not simply due to *having* a verifier, we apply CAMS’s own post-processor—the two-direction M_{ver} check, NLI citation re-retrieval over a common span pool, and bounded drop (§3.6, $R=2$)—uniformly to every system (Appendix D, Table 9). The same verifier helps the baselines far less: it raises end-to-end precision only $59 \rightarrow 64$ and Attribute First $76 \rightarrow 77$, each at a sharp coverage cost (e.g. $.66 \rightarrow .49$) as unsupported sentences are dropped rather than repaired. Applied to the verifier-ablated CAMS variant it instead recovers the full pipeline’s precision ($77 \rightarrow 84$) at unchanged recall, reproducing CAMS (full). Since the post-processor is shared, the difference comes from what it operates on: CAMS feeds it span-anchored, localized claims whose citation failures are usually repairable, whereas baseline sentences more often must be dropped.

5.4 Localization Accuracy (Regime 2)

Table 3 isolates localization from generation using oracle content. On overall span-F1, CAMS is on par with the span-first method Attribute First (77 vs 72), confirming that claim anchoring does not sacrifice single-source localization. The decisive difference is on *multi-source recall*: for propositions whose gold support is distributed across two or more documents, CAMS attributes correctly to multiple sources 64% of the time versus 38% for Attribute First, with non-overlapping 95% confidence intervals. Ablating cross-document clustering drops this figure to 31%, below even the span-first baseline (38%): without clustering each atomic claim stays tied to one document’s span, so distributed facts are attributed to a single source, whereas the span-first baseline can incidentally select spans from several documents. Cross-document merging is thus not a bonus but the mechanism that makes the claim representation competitive on multi-source attribution. The retrieval lower bound and gold-claim ceiling bracket these numbers and separate matching error from localization ability.

System	Span P	Span R	Span F1	Multi Src
Retrieval (LB)	41	55	47	22
Attribute First	74	70	72	38
CAMS (ours)	76	78	77	64
– clustering	75	71	73	31
Gold-claim (UB)	89	92	90	81

Table 3: Regime 2 localization at $\tau = 0.5$. Multi-Src is multi-source recall; LB/UB are the retrieval lower bound and gold-claim ceiling. τ -sensitivity and bootstrap CIs are in Appendix E.

System	Diverg. cov.	Conflict surf.	Conflict attr. acc.
E2E LLM (ALCE)	41	18	–
Attribute First	47	22	55
CAMS (ours)	58	61	73

Table 4: Conflict and diversity on DiverseSumm: coverage of divergent facts, fraction of inter-source conflicts surfaced, and per-source conflict attribution accuracy.

5.5 Conflict and Diversity (DiverseSumm)

On DiverseSumm, which targets diverse and conflicting information across sources, CAMS covers 58% of annotated divergent facts and surfaces 61% of inter-source conflicts, attributing each side to its originating documents with 73% accuracy (Table 4), whereas the end-to-end model surfaces only 18%, typically reporting a single consensus value. Because conflict detection operates over normalized claim clusters, disagreement becomes an explicit, attributable object rather than something the decoder silently resolves.

5.6 Zero-Shot Transfer to WCEP

To test whether the framework depends on in-domain selector training, we apply the MultiNews-trained selector to WCEP without WCEP-specific training or tuning. Table 5 shows that CAMS remains competitive in quality and retains its faithfulness advantage, although Attribute First obtains slightly higher ROUGE-L. This supports the view that claim anchoring and verification transfer more robustly than salience selection.

5.7 Controllable Faithfulness–Coverage Trade-off

Sweeping the selection threshold θ traces a monotonic faithfulness–coverage frontier for CAMS that lies above the baselines under the reported metrics—at comparable coverage it scores higher on measured faithfulness, and vice versa (Figure 2, Ap-

System	R-L	BERT Sc.	Align Sc.	Sum maC	FAct Sc.
E2E LLM (ALCE)	30.5	86.8	.77	.69	.73
Attribute First	31.2	87.1	.83	.77	.80
CAMS (ours)	30.9	87.5	.88	.82	.89
– verification	30.8	87.3	.84	.79	.83

Table 5: Zero-shot transfer to WCEP. The selector is trained only on MultiNews and applied to WCEP without WCEP-specific training or tuning. Only the WCEP test split is used.

pendix E). End-to-end models occupy a single fixed point off this frontier, so their balance is not steerable, whereas in CAMS it is an explicit selector knob.

5.8 Human Evaluation and Metric Validity

Annotators judge 89% of CAMS claims fully supported and verify each in 9s on average—about $3.4\times$ faster than for the document-citing baseline—because span-level, concise citations point straight to the evidence (Appendix D, Table 10). Human support correlates with independent citation precision at $r = 0.78$, recall at $r = 0.71$, and F1 at $r = 0.75$ (agreement $\kappa = 0.68$); the precision correlation is the most relevant guard against circularity, since CAMS is explicitly verified before evaluation.

5.9 Ablations

Our ablations isolate each module. Removing verification lowers faithfulness (Table 1); removing cross-document clustering collapses multi-source recall while leaving single-source localization intact (Table 3). A low-resource study (Appendix E, Table 12) shows the selector trained on as few as 1k examples retains 92% of full multi-source recall and nearly all faithfulness, echoing the modular advantage of ESR (Guan and Padmakumar, 2023): components stay effective and independently trainable when data are scarce. Intrinsic per-module diagnostics (extraction $90 \rightarrow 94\%$ claim support, 97% quote-match rate, clustering B³ F1 84, conflict-detection F1 72) appear in Appendix F.

6 Conclusion

We presented CAMS, a claim-anchored modular framework for multi-document summarization in which atomic claims with provenance serve as both verification unit and attribution anchor. By localizing content before realizing it, CAMS is

attribution-oriented by construction—every sentence traces to one or more source spans—and faithfulness-oriented, though not faithful, by construction. Empirically it matches end-to-end quality while improving faithfulness and, most distinctively, raising correct attribution of multi-source evidence from 38% to 64% over span-first baselines (a 68% relative gain), and it transfers zero-shot to WCEP with only a modest drop. We hope the claim-anchored view helps reconnect attribution and atomic-fact factuality, usually studied apart.

Limitations

CAMS depends on the quality of LLM-based claim decomposition; extraction errors propagate downstream, decontextualization can supply an incorrect entity, time, or location, and claim *equivalence* for clustering is inherently fuzzy, so clustering errors can mis-merge or split evidence. Attribution is many-to-many—a sentence may aggregate several claims from several documents—and evaluating such distributed attribution against gold is an open problem that our soft-matching protocol only approximates. Although we add an evaluator-decoupled audit, citation evaluation still relies on automatic support models, and no automatic metric fully substitutes for human judgments; the independent evaluator should therefore be read as a robustness check, not as proof that circularity or verifier error is impossible. Similarly, quote-to-span matching establishes where a quoted string occurs, not by itself that the normalized claim preserves exactly the same meaning. The pipeline incurs extraction cost that grows with the number and length of source documents, raising latency relative to end-to-end models. Our conflict-handling policy makes a design choice (surface vs. defer) that may not suit all applications. Finally, our experiments are limited to English news; generalization to other domains and languages remains untested.

Ethics Statement

Summarization systems can mislead if they distort or omit source content; by making every statement traceable to source spans, CAMS is intended to support, not replace, human verification. Attribution should not be read as a guarantee of real-world truth, only of support by the provided sources, which may themselves be unreliable. We use publicly available datasets consistent with their intended research use and report compute in Ap-

pendix A.

References

- Gabor Angeli, Melvin Jose Johnson Premkumar, and Christopher D. Manning. 2015. Leveraging linguistic structure for open domain information extraction. In *Proceedings of ACL-IJCNLP*.
- Iz Beltagy, Matthew E. Peters, and Arman Cohan. 2020. Longformer: The long-document transformer. In *arXiv preprint arXiv:2004.05150*.
- Bernd Bohnet, Vinh Q. Tran, Pat Verga, Roei Aharoni, Daniel Andor, Livio Baldini Soares, Jacob Eisenstein, Jonathan Herzig, Kelvin Gaitonde, Prateek Jain, and 1 others. 2022. Attributed question answering: Evaluation and modeling for attributed large language models. In *Proceedings of EMNLP*.
- Jaime Carbonell and Jade Goldstein. 1998. The use of mmr, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of SIGIR*.
- Eunsol Choi, Jennimaria Palomaki, Matthew Lamm, Tom Kwiatkowski, Dipanjan Das, and Michael Collins. 2021. Decontextualization: Making sentences stand-alone. *Transactions of the Association for Computational Linguistics*, 9:447–461.
- Ori Ernst, Mohit Bansal, Noah A. Smith, and Greg Durrett. 2021. A human-aligned span-level evaluation framework for text summarization. In *Proceedings of ACL-IJCNLP*.
- Alexander R. Fabbri, Wojciech Kryscinski, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir R. Radev. 2022. Qafacteval: Improved qa-based factual consistency evaluation for summarization. In *Proceedings of NAACL*.
- Alexander R. Fabbri, Irene Li, Tianwei She, Suyi Li, and Dragomir R. Radev. 2019. Multi-news: A large-scale multi-document summarization dataset and abstractive hierarchical model. In *Proceedings of ACL*.
- Jerome H. Friedman. 2001. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of EMNLP*.
- Demian Gholipour Ghalandari, Chris Hokamp, Nghia The Pham, John Glover, and Georgiana Ifrim. 2020. [A large-scale multi-document summarization dataset from the wikipedia current events portal](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1302–1308, Online. Association for Computational Linguistics.

- Tanya Goyal and Greg Durrett. 2022. News summarization and evaluation in the era of gpt-3. *arXiv preprint arXiv:2209.12356*.
- Shuo Guan. 2021. [Knowledge and keywords augmented abstractive sentence summarization](#). In *Proceedings of the Third Workshop on New Frontiers in Summarization*, pages 25–32, Online and in Dominican Republic. Association for Computational Linguistics.
- Shuo Guan and Vishakh Padmakumar. 2023. [Extract, select and rewrite: A modular sentence summarization method](#). In *Proceedings of the 4th New Frontiers in Summarization Workshop*, pages 41–48, Singapore. Association for Computational Linguistics.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. In *Proceedings of ICLR*.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansky, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. True: Re-evaluating factual consistency evaluation. In *Proceedings of NAACL*.
- Kung-Hsiang Huang, Philippe Laban, Alexander R. Fabbri, Prafulla Kumar Choubey, Shafiq Joty, Caiming Xiong, and Chien-Sheng Wu. 2024. [Embrace divergence for richer insights: A multi-document summarization benchmark and a case study on summarizing diverse information from news articles](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 570–593, Mexico City, Mexico. Association for Computational Linguistics.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Wojciech Kryscinski, Bryan McCann, Caiming Xiong, and Richard Socher. 2020. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of EMNLP*.
- Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. 2022. Summac: Re-visiting nli-based models for inconsistency detection in summarization. In *Transactions of the Association for Computational Linguistics*.
- Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. 2023. Towards general text embeddings with multi-stage contrastive learning. *arXiv preprint arXiv:2308.03281*.
- Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of EMNLP*.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of ACL*.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of EMNLP*.
- Mike Mintz, Steven Bills, Rion Snow, and Daniel Jurafsky. 2009. Distant supervision for relation extraction without labeled data. In *Proceedings of ACL-IJCNLP*.
- Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and Ehud Reiter. 2023. Measuring attribution in natural language generation models.
- Aviv Slobodkin, Eran Hirsch, Arie Cattan, Tal Schuster, and Ido Dagan. 2024. [Attribute first, then generate: Locally-attributable grounded text generation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3309–3344, Bangkok, Thailand. Association for Computational Linguistics.
- Guangfen Wei, Gang Li, Shuo Guan, Jie Zhao, and Xue Sun. 2018. [Study on an improved lenet-5 gas identification structure for electronic noses](#). In *2018 IEEE SENSORS*, pages 1–4. IEEE.
- Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL*.
- Wen Xiao, Iz Beltagy, Giuseppe Carenini, and Arman Cohan. 2022. Primera: Pyramid-based masked sentence pre-training for multi-document summarization. In *Proceedings of ACL*.
- Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. Alignscore: Evaluating factual consistency with a unified alignment function. In *Proceedings of ACL*.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert. In *Proceedings of ICLR*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *ArXiv preprint arXiv:2306.05685*.

A Dataset Statistics and Compute

Table 6 summarizes the datasets and their role in the evaluation. MultiNews is used as the primary benchmark for quality and faithfulness; DiverseSumm is used for diversity and conflict analysis; WCEP is used as a zero-shot transfer setting in which the selector receives no WCEP-specific training—only its test split is evaluated (§5.6). All systems are evaluated on the same preprocessed inputs and matched output-length constraints.

Dataset	Train	Dev	Test	Primary use
MultiNews	44,972	5,622	5,622	Main MDS benchmark
DiverseSumm	—	—	245	Conflict/diversity analysis
WCEP	~8.2k	~1.0k	~1.0k	Zero-shot transfer

Table 6: Dataset statistics used in the experiments. DiverseSumm contains 245 stories with 10 documents per story; WCEP uses the public chronological split. WCEP train/dev sizes are listed for completeness only—they are not used, as WCEP is a zero-shot transfer setting (§5.6).

Implementation and compute. All systems use the same document preprocessing, maximum input budget, and output-length constraints. Claim extraction, fact decomposition, and rewriting are performed with the frozen API model of §4.4; embeddings are computed once, ℓ_2 -normalized, and cached for clustering. The selector is trained on the MultiNews training split and tuned on development data; the final operating point is $\theta^*=0.6$, selected by maximizing attribution F1 subject to a minimum coverage constraint. The full run uses four A100 GPUs, with extraction, clustering, selection, rewriting, and verification logged separately for reproducibility.

B Reproducibility Configuration

Table 7 lists the concrete hyperparameters used in the main experiments. The independent evaluator M_{eval} is intentionally separated from M_{sel} and M_{ver} and is never called during selection, generation, verification, repair, or threshold tuning.

C Algorithms and Module Details

Extraction schema. The extractor returns a JSON array; each element has fields `claim` (a decontextualized proposition), `quote` (a verbatim substring of the source that licenses the claim), and `doc_id`. The prompt forbids reporting character or

Algorithm 1 Quote-to-span provenance resolution

Require: document d , verbatim quote q , similarity floor ρ , tokenizer offsets `OFF`
Ensure: token span (start, end) or `NONE`

```

1:  $(a, b) \leftarrow \text{EXACTSUBSTR}(d, q)$   $\triangleright$  char interval
2: if  $(a, b) = \text{NONE}$  then
3:    $\tilde{d}, \tilde{q} \leftarrow \text{NORM}(d), \text{NORM}(q)$   $\triangleright$  case/space/punct
4:    $(a, b), \sigma \leftarrow \text{FUZZYWINDOW}(\tilde{d}, \tilde{q})$   $\triangleright$  indel sim.
5:   if  $\sigma < \rho$  then return NONE
6:    $(a, b) \leftarrow \text{MAPTORAW}(a, b)$   $\triangleright$  undo NORM
7: return CHARTOTOK( $a, b, \text{OFF}(d)$ )

```

token offsets and requires the quote to be copied verbatim; offsets are recovered post hoc by Algorithm 1. Decontextualization—pronoun resolution and restoration of elided subject, time, and place (Choi et al., 2021)—is requested in the same instruction, while the separate verbatim quote field preserves an unedited string for localization.

Provenance resolution. Algorithm 1 gives the three-tier matcher. Normalization lower-cases text, collapses whitespace, and standardizes quotation marks and dashes; the fuzzy stage scores candidate windows of d by indel (normalized Levenshtein) similarity and accepts the best window above $\rho=0.85$. Indices in the normalized string are mapped back to raw character offsets, then to token indices via the tokenizer’s `offset_mapping`. For documents exceeding the context window we extract over ~ 512 -token sliding windows (64-token overlap) and add each chunk’s global character offset before resolution; overlap duplicates are removed during clustering (§3.3).

Selector training (distant supervision). We build silver labels by aligning each gold reference sentence to the clusters: a cluster is positive if some reference sentence is entailed by its claim under M_{sel} (backing off to embedding/ROUGE similarity above a fixed cutoff when entailment is borderline), and negative otherwise (Mintz et al., 2009). The LightGBM scorer (Ke et al., 2017) is trained on these labels; only salience and self-support are pointwise features, while coverage is computed online during the greedy loop of Algorithm 2. M_{eval} is held out completely from this process.

Rewriter and verifier. The rewriter receives cluster-labeled claims (“[g_j] ...”) and must end each sentence with a citation marker such as [g_3, g_7]; markers are parsed and expanded to source spans through $g_j \mapsto \text{SPANS}(g_j)$. The verifier runs the two-direction check of §3.6: M_{ver} support of

Component	Setting
LLM API	claude-opus-4-8; calls run from 2026-05-28 to 2026-06-03; three few-shot examples; max output tokens: extraction 4,096, rewrite 768, fact decomposition 1,024; temperature/top- p omitted
Preprocessing	512-token chunks; 64-token overlap; exact quote matching attempted before fuzzy matching; fuzzy matcher uses rapidfuzz indel similarity with acceptance $\rho=0.85$
Embeddings/clustering	gte-large; normalized embeddings; cosine similarity; agglomerative clustering with average linkage; similarity threshold 0.82 / distance threshold 0.18; merge check by bidirectional entailment
NLI modules	Conflict model: DeBERTa-v3-MNLI three-class NLI, contradiction threshold 0.7; support thresholds tuned over $\{0.5, 0.7\}$; M_{sel} , M_{ver} , and M_{eval} are separate models
Selector	LightGBM GBDT, binary objective, random seed 42, 31 leaves, learning rate $\{0.05, 0.1\}$, 100–500 estimators, early stopping 50 rounds, MMR $\lambda=0.5$, $\theta \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$, main $\theta^*=0.6$
Verification	At most two repair retries; repair order: citation repair, regeneration, then drop unsupported content
Evaluation	MultiNews full test split 5,622 unless a 500/1,000-example diagnostic sample is explicitly marked; DiverseSumm 245; WCEP roughly 1,000; main localization $\tau=0.5$, sensitivity $\tau \in \{0.3, 0.5, 0.7\}$; 1,000 paired-bootstrap samples; 95% percentile confidence intervals; significance by paired-bootstrap $p < 0.05$; human audit uses 100–300 claims per system, three annotators, Cohen’s/Fleiss’ κ

Table 7: Reproducibility configuration for the main experiments.

Algorithm 2 Greedy support-aware selection (MMR)

Require: clusters G , pointwise scorer s_{GBDT} , threshold θ , weight λ
Ensure: ordered selection S
1: $S \leftarrow []$
2: **loop**
3: **for** $g \in G \setminus S$ **do**
4: $r(g) \leftarrow s_{\text{GBDT}}(g) - \lambda \max_{g' \in S} \text{sim}(g, g')$ ▷
 $\max_{\emptyset}=0$
5: $g^* \leftarrow \arg \max_g r(g)$
6: **if** $r(g^*) < \theta$ **then break**
7: append g^* to S
8: **return** S

the sentence by its cited spans, and FActScore-style per-atomic-fact entailment to detect overflow. Citation-only failures trigger NLI re-retrieval (citation repair) before any regeneration, with at most $R=2$ retries. M_{eval} is never queried in this loop.

D Evaluator-Decoupled Robustness

Table 8 isolates how much of the citation result depends on model choice. The internal verifier score is included only for diagnosis; the independent evaluator is the model used for the main Regime 1 precision claim. Removing self-support from the selector lowers precision even under the independent evaluator, and removing verification lowers it more substantially. This pattern is inconsistent with a pure circularity explanation in which gains appear only under the model used for filtering.

System	Prec (M_{eval})	TRUE Prec diag.	Human Supp.
CAMS (ours)	84	88	89
– self-support feat.	81	84	86
– verification	77	82	83
Attribute First	76	78	82

Table 8: Evaluator-decoupled robustness on MultiNews. M_{eval} is never used during generation, selection, verification, repair, or threshold tuning; TRUE precision is reported as a diagnostic only.

E Regime 2 Bounds and Sensitivity

Table 11 reports the localization results under different soft-overlap thresholds. The ranking is stable across thresholds: stricter matching lowers absolute F1 for all systems, but CAMS retains the largest advantage on multi-source propositions.

For significance testing we use paired bootstrap resampling over document clusters with 1,000 samples and report 95% percentile confidence intervals for automatic metrics. The main tables report the development-selected operating point, $\theta^*=0.6$. The retrieval lower bound uses the same candidate span pool without claim clustering, while the gold-claim upper bound supplies oracle normalized claims before localization. These bounds separate retrieval failure from errors introduced by clustering, selection, and rewriting.

System	Cite P		Cite R		FAct Sc.	Cov.
	raw	post	raw	post		
E2E (ALCE)	59	64	70	53	.75 $\rightarrow$.79	.66 $\rightarrow$.49
Attr. First	76	77	78	63	.82 $\rightarrow$.83	.58 $\rightarrow$.53
CAMS – verif.	77	84	82	81	.86	.62
CAMS (full)	84	84	81	81	.90	.60

Table 9: Fair post-processing control on MultiNews (§5.3). Every system’s raw output is passed through CAMS’s shared verifier: the two-direction M_{ver} check, NLI citation re-retrieval over a common span pool, and bounded drop ($R=2$, §3.6). The “raw” Cite columns coincide with the Regime 1 numbers (Table 2); the verifier-ablated row matches the ablation in Table 8. The same post-processor raises baseline precision only slightly and at a large coverage cost, while reconstructing CAMS’s full-pipeline precision (84) from its verifier-ablated variant. FActScore/coverage for the CAMS rows are operating-point values.

System	% Supp.	Verify time (s)
E2E LLM (ALCE)	68	31
Attribute First	82	14
CAMS (ours)	89	9

Table 10: Human evaluation (§5.8): % of claims judged fully supported and mean verification time per claim. Human support correlates with independent citation precision ($r = 0.78$), recall ($r = 0.71$), and F1 ($r = 0.75$); inter-annotator agreement $\kappa = 0.68$.

F Per-Module Diagnostics

Tables 13–16 give the intrinsic accuracy of the four upstream stages (claim extraction, quote-to-span matching, cross-document clustering, and conflict detection; §3.2–§3.3), measured on held-out diagnostic samples; all values are percentages. These diagnostics localize where end-to-end errors originate: extraction and decontextualization bound downstream faithfulness, quote matching bounds localization accuracy, clustering governs multi-source attribution, and conflict detection governs what disagreement can be surfaced.

τ	Attribute First	CAMS	Δ
0.3	78	83	+5
0.5	72	77	+5
0.7	61	68	+7

Table 11: Span-F1 sensitivity under different token-IoU thresholds.

Selector data	Span-F1	Multi-Src	FActSc.
full	77	64	.90
5k	76	62	.89
1k	74	59	.87

Table 12: Low-resource ablation on the selector (§5.9): performance degrades gracefully as training data shrink.

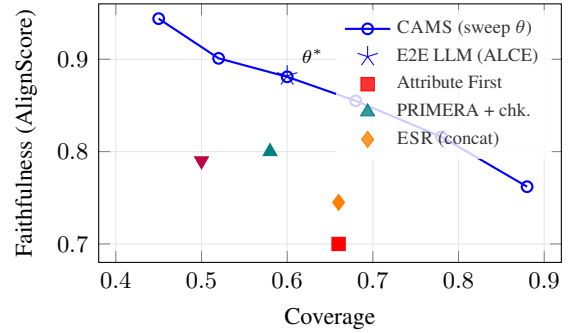


Figure 2: Measured faithfulness–coverage trade-off controlled by the selection threshold θ (§5.7). CAMS sweeps a frontier (operating point θ^*) that lies above the baselines under the reported faithfulness-oriented metrics, each baseline being a single fixed point.

Metric	Value
JSON schema-valid outputs	99
Claims per source document	21
Quote-localizable rate	97
Claim supported by quote (raw)	90
after self-support filtering	94
Atomicity (human/LLM)	90
Salient reference-fact recall	81
Decontextualization/hallucination error	12

Table 13: Claim extraction diagnostics. “Supported” is the claim being entailed by its own verbatim quote, before and after the selector’s self-support filtering (§3.4); salient-fact recall is against reference-salient facts, not all source facts.

Metric	Value
Exact-match rate	88
Normalized/fuzzy rescue	+12
Final matched rate	97
Wrong-occurrence rate	4
Token-span IoU ≥ 0.5	94
Token-span exact-boundary accuracy	88
Claims dropped (unmatched quote)	3

Table 14: Quote-to-span matching diagnostics. Normalized fuzzy rescue adds 12 points over exact matching (Algorithm 1).

Metric	Value
Pairwise precision	90
Pairwise recall	78
B ³ / cluster F1	84
Over-merge rate	3
Under-merge / split rate	21
Multi-source evidence cluster recall	70
Pairwise precision (no NLI merge check)	71
Pairwise recall (no NLI merge check)	73

Table 15: Cross-document clustering diagnostics. The bidirectional-entailment merge check raises pairwise precision from 71 to 90 at a small recall cost (§3.3).

Metric	Value
Candidate-pair recall	85
Candidate-pair precision	21
NLI contradiction precision	81
NLI contradiction recall	65
Final conflict-detection F1	72
Conflict surfaced after select/rewrite	65
Conflict attribution accuracy	71

Table 16: Conflict-detection diagnostics. The cheap entity/number candidate generator is high-recall, low-precision by design; the three-class NLI filter supplies precision (§3.3).